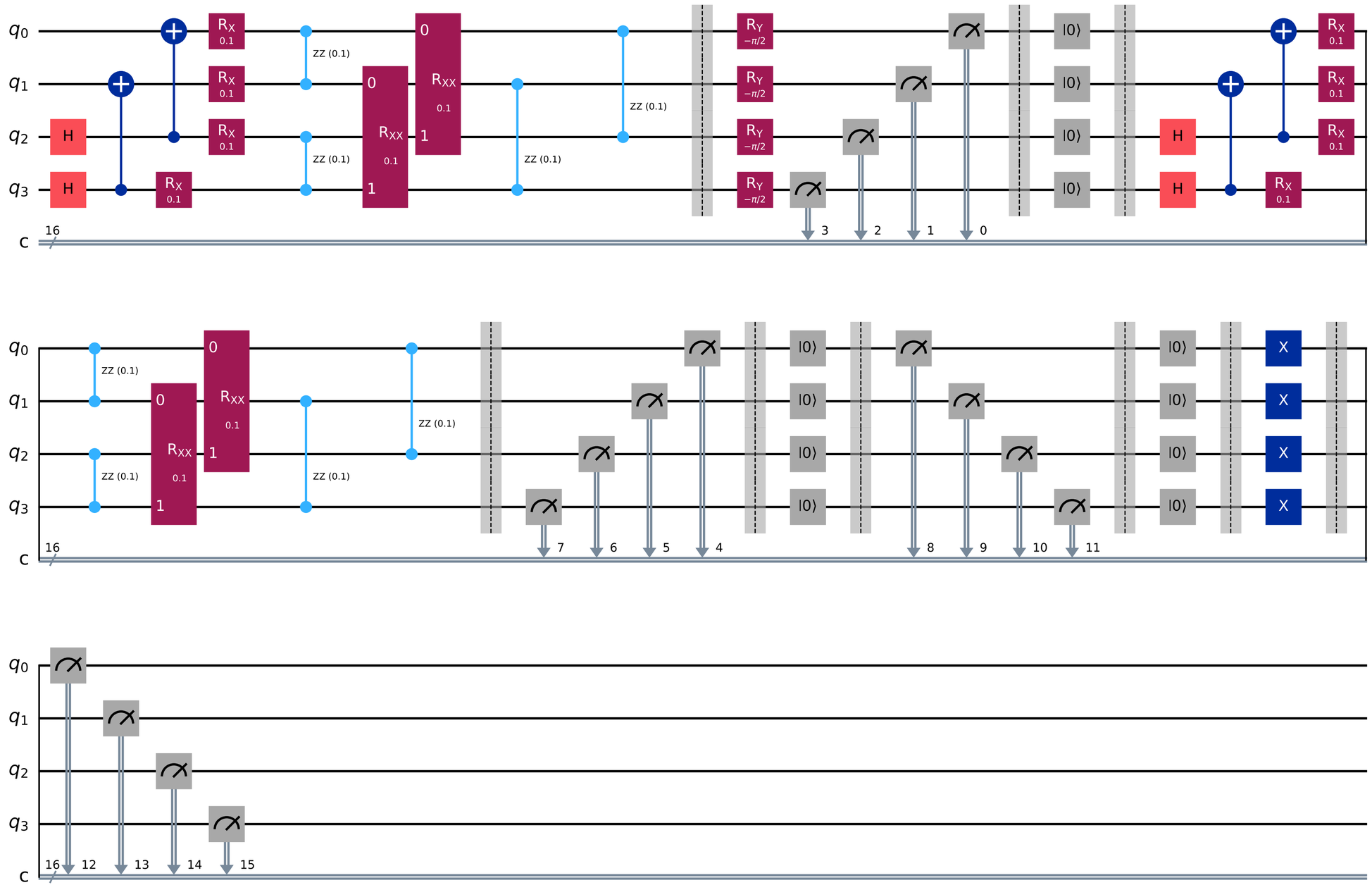
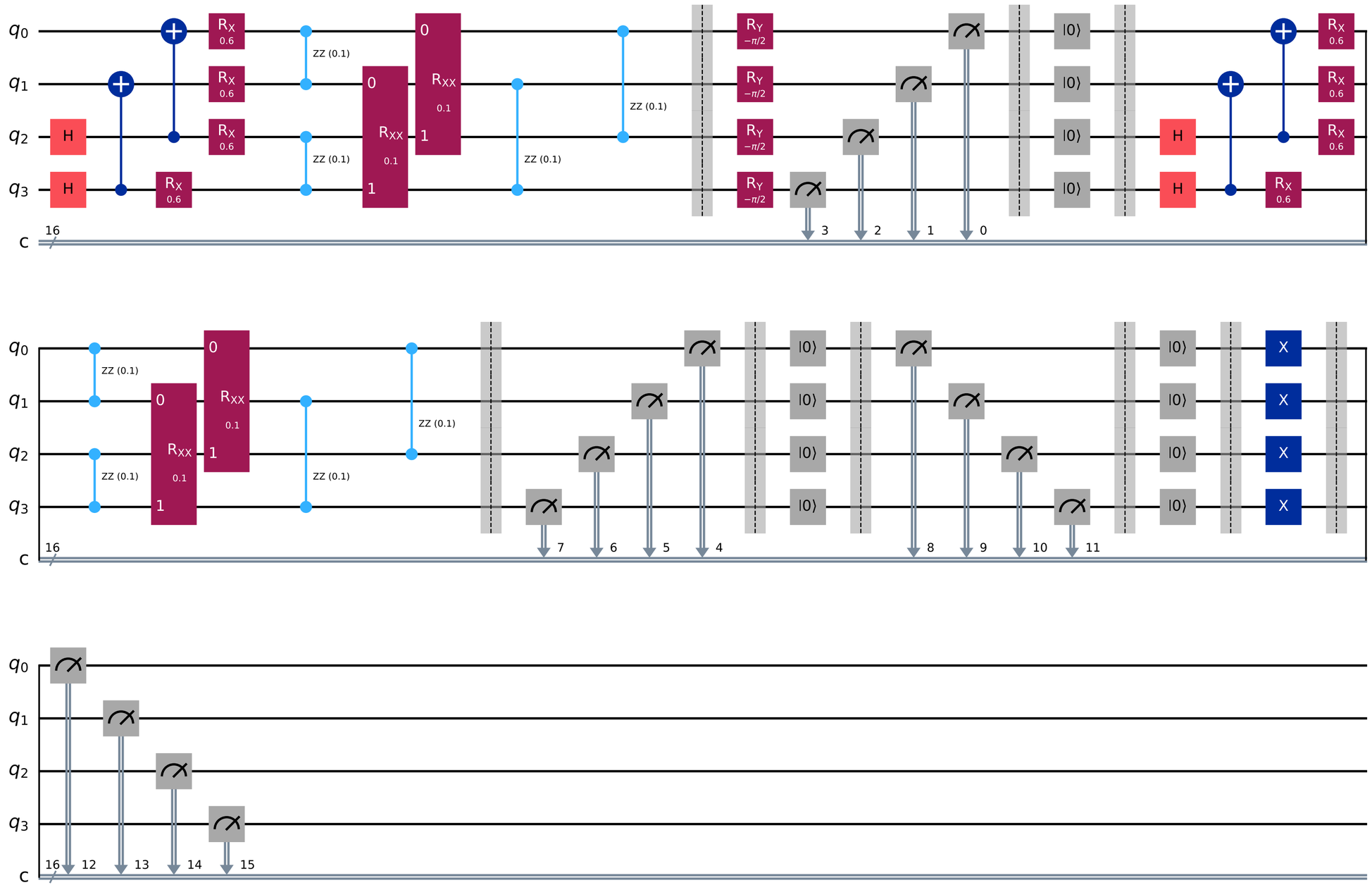
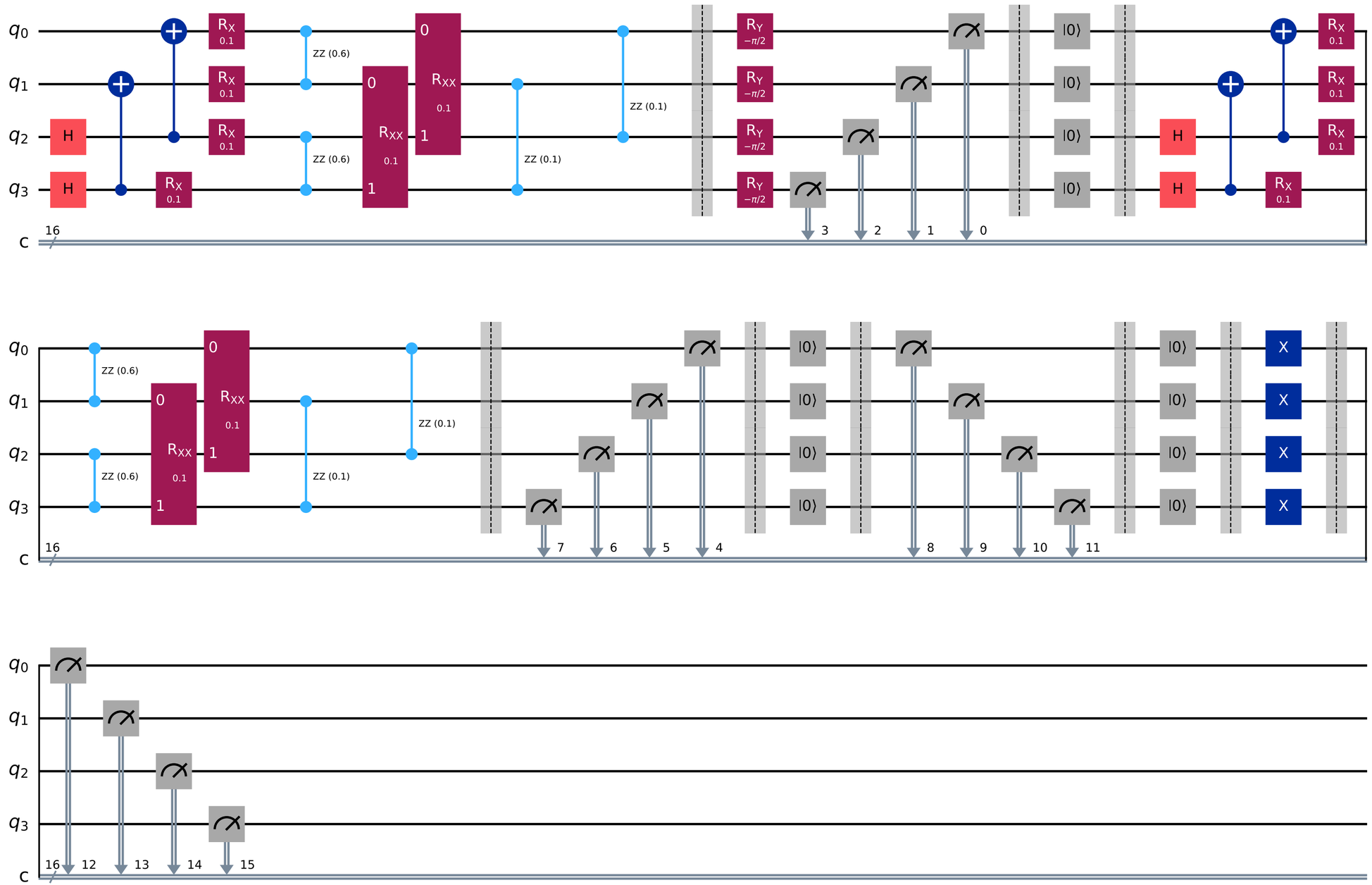
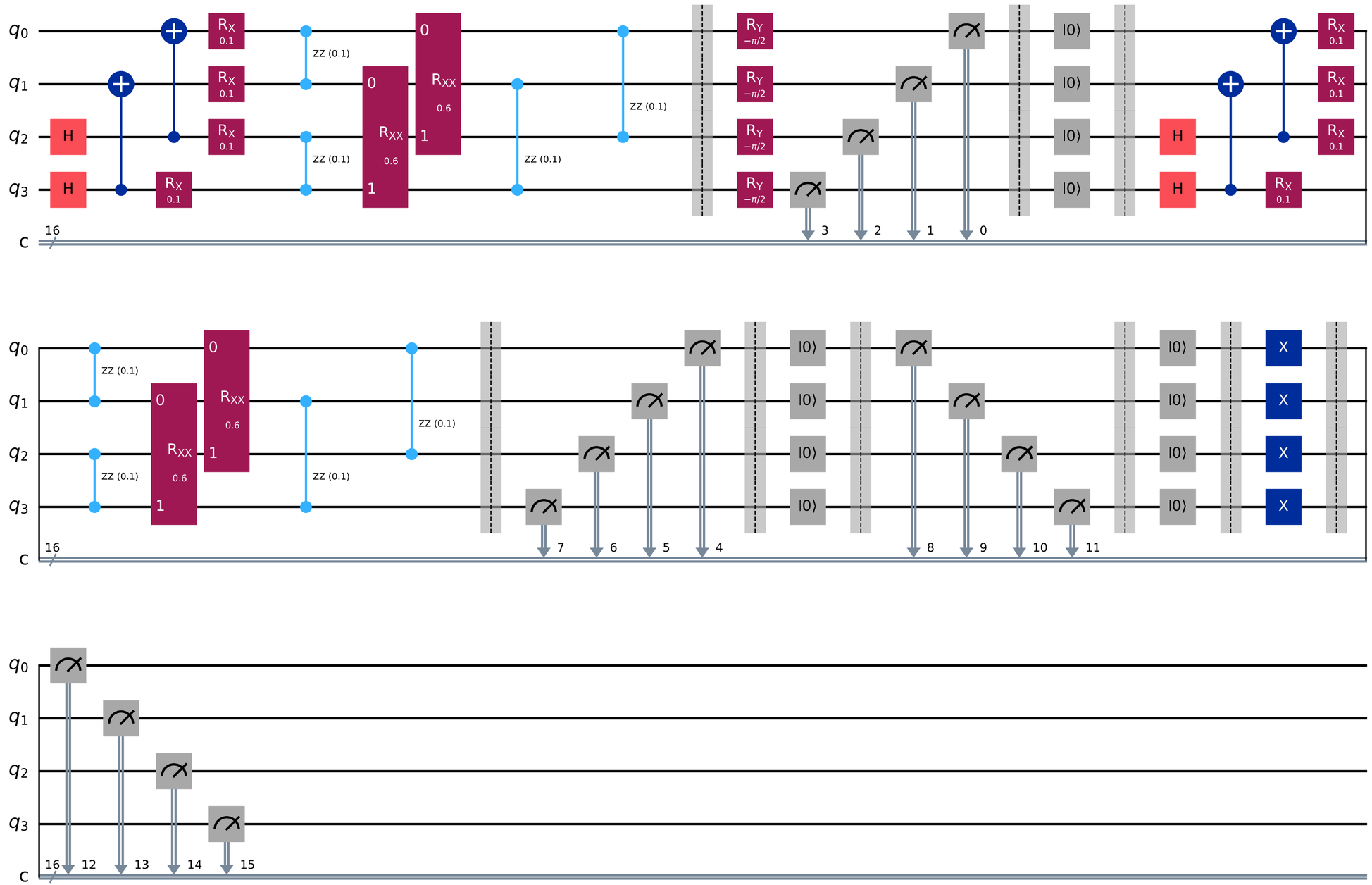
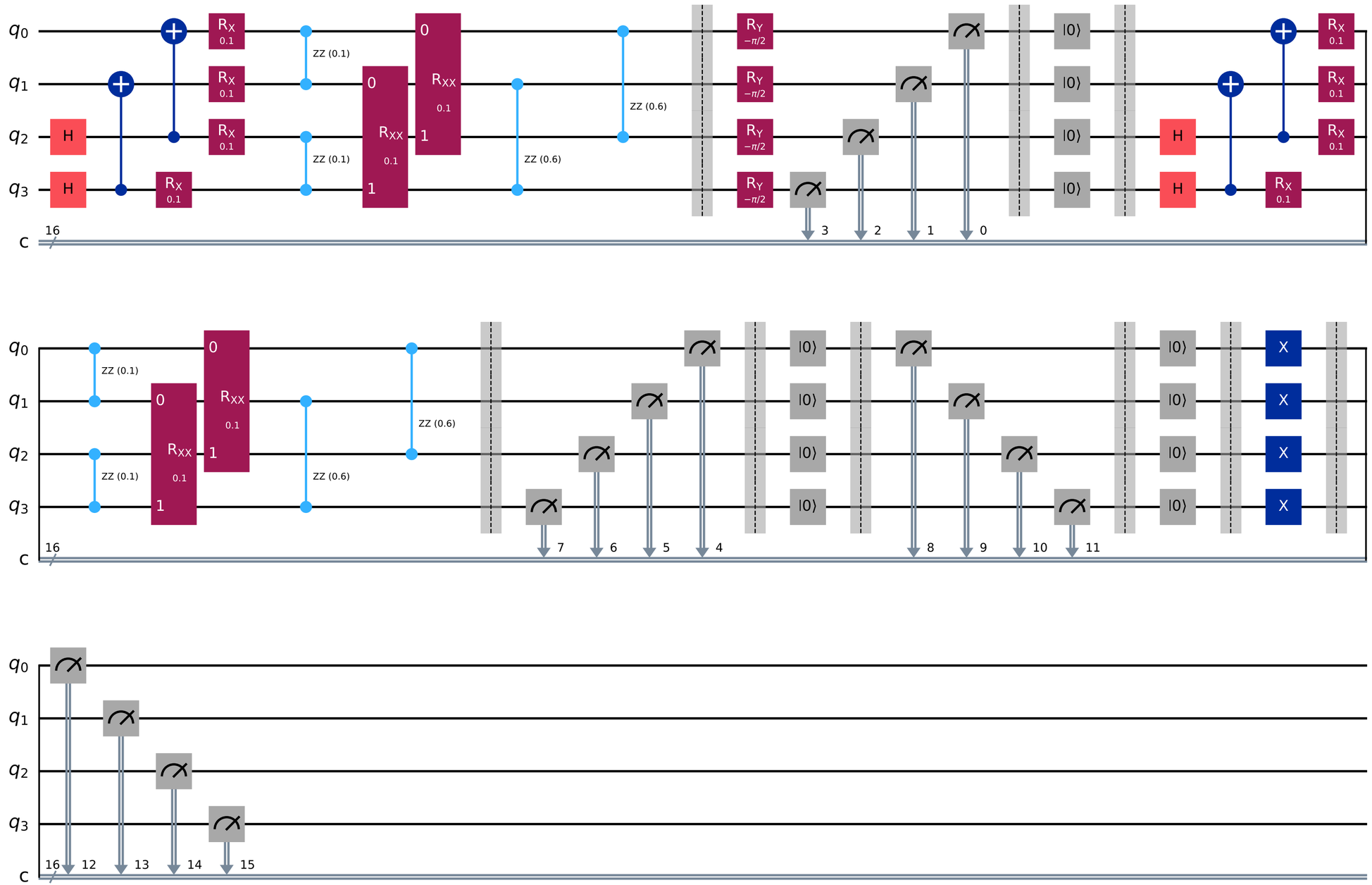
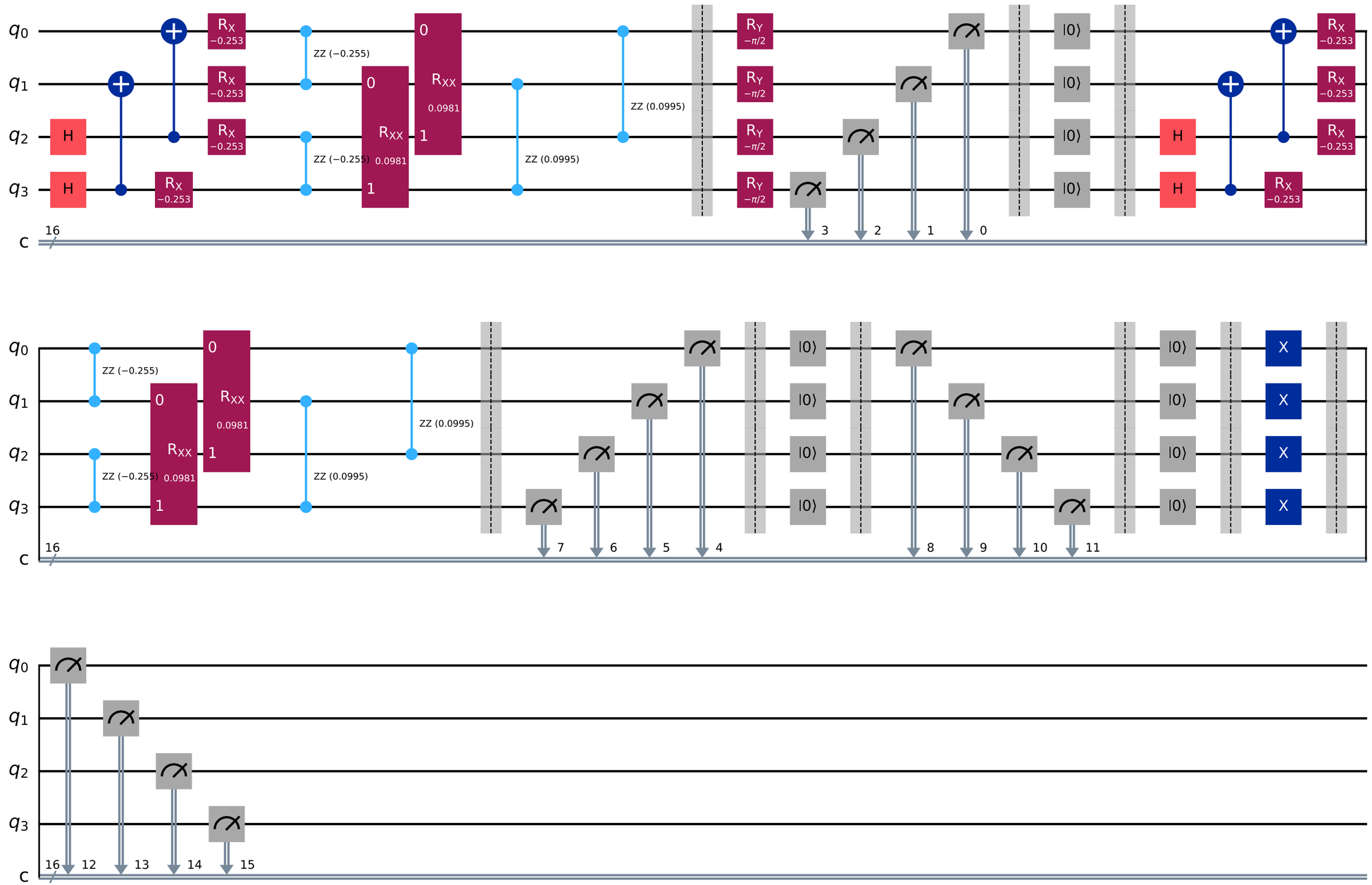

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

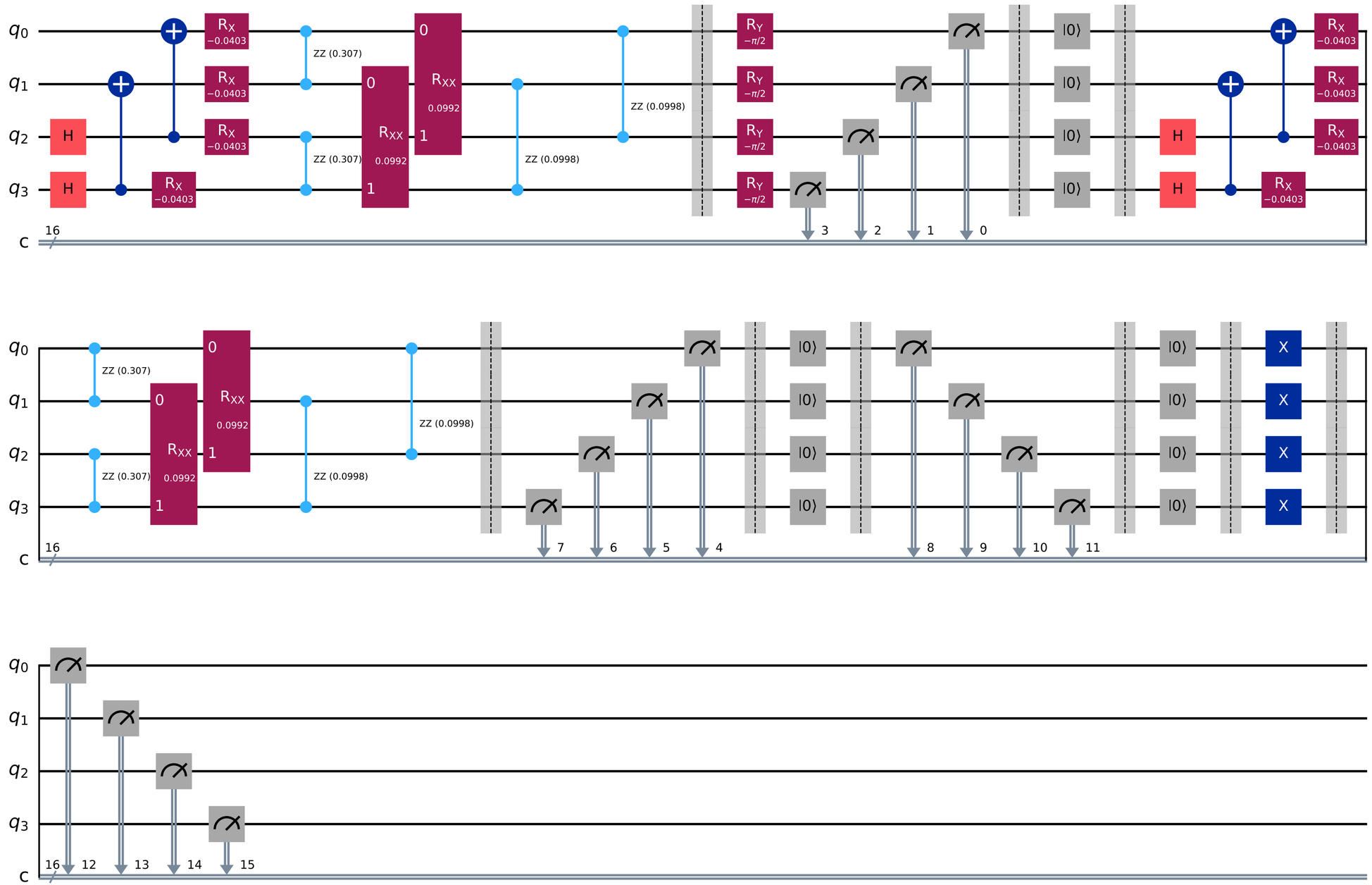

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.6
R_{XX} 0.1
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

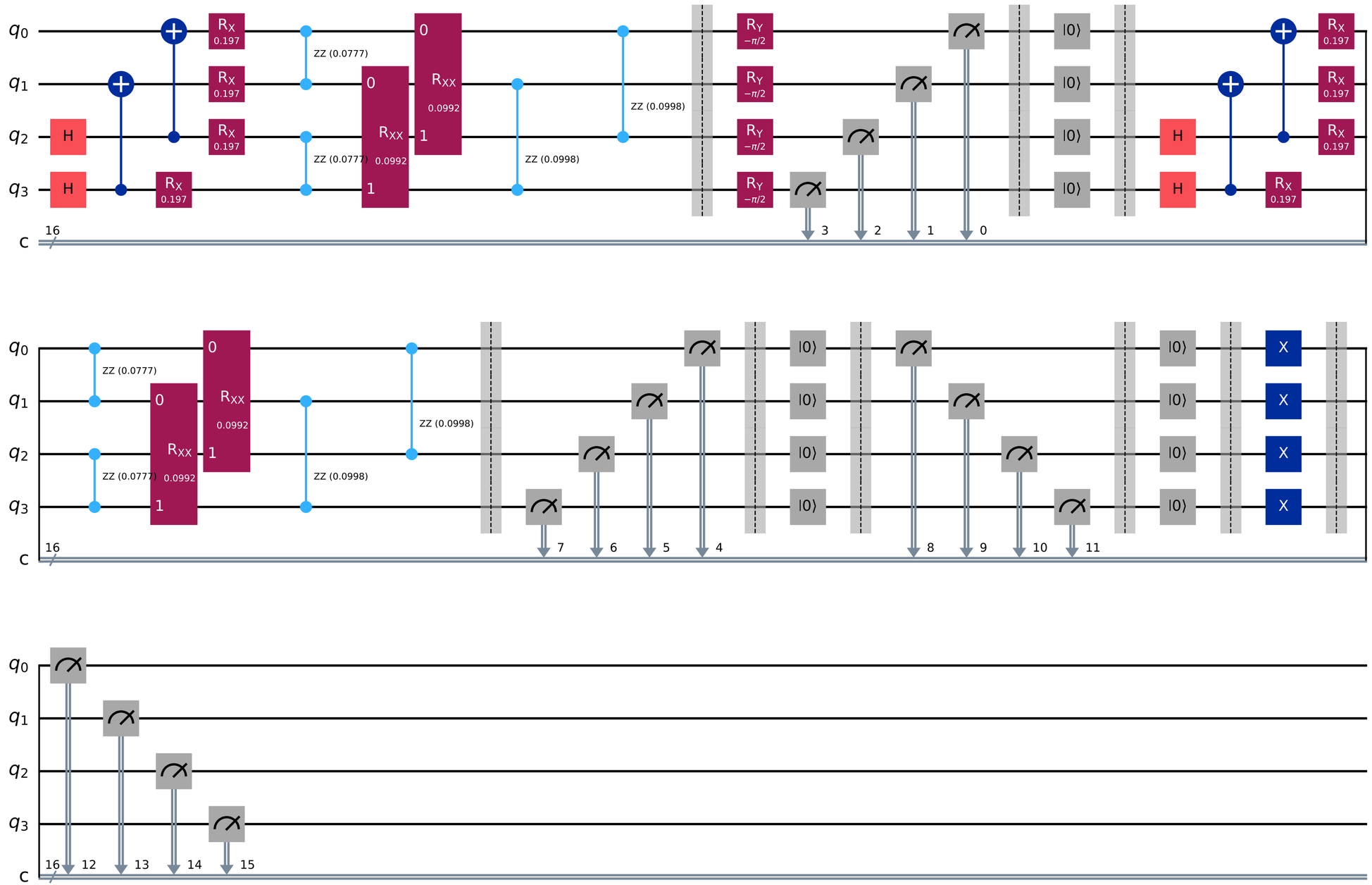

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.6)
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

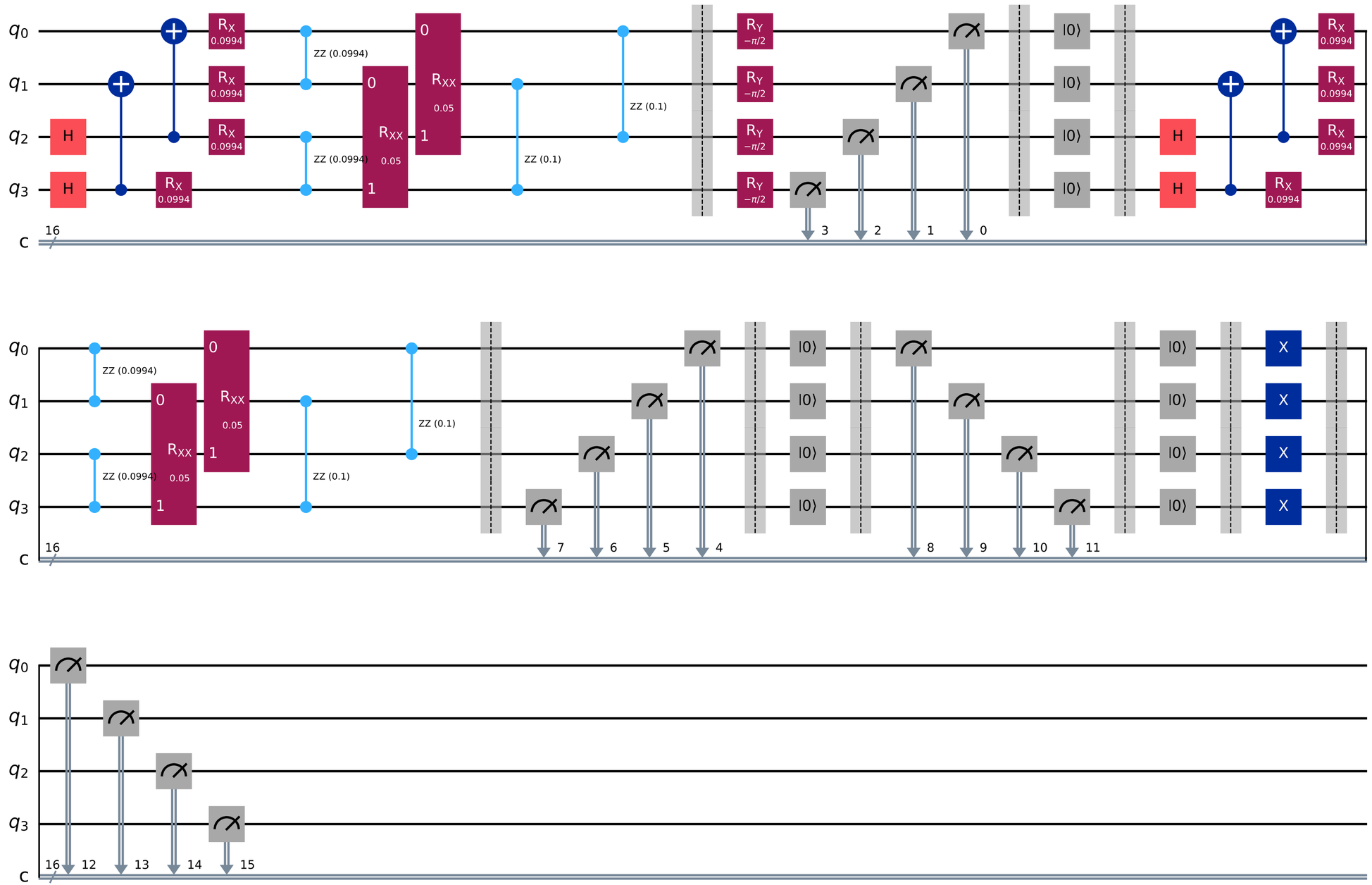

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.6
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
0
1
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

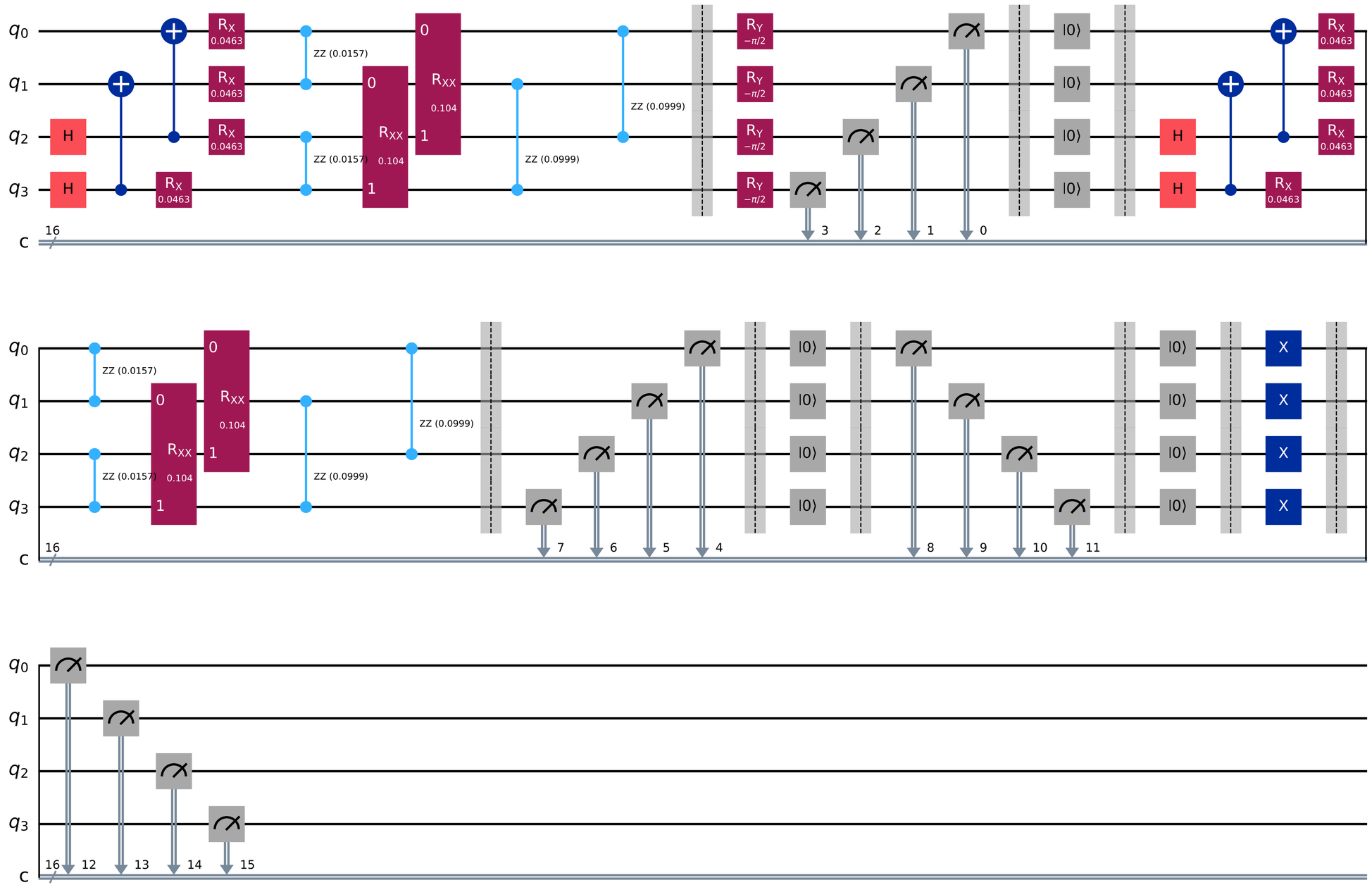

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.1)
ZZ (0.6)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

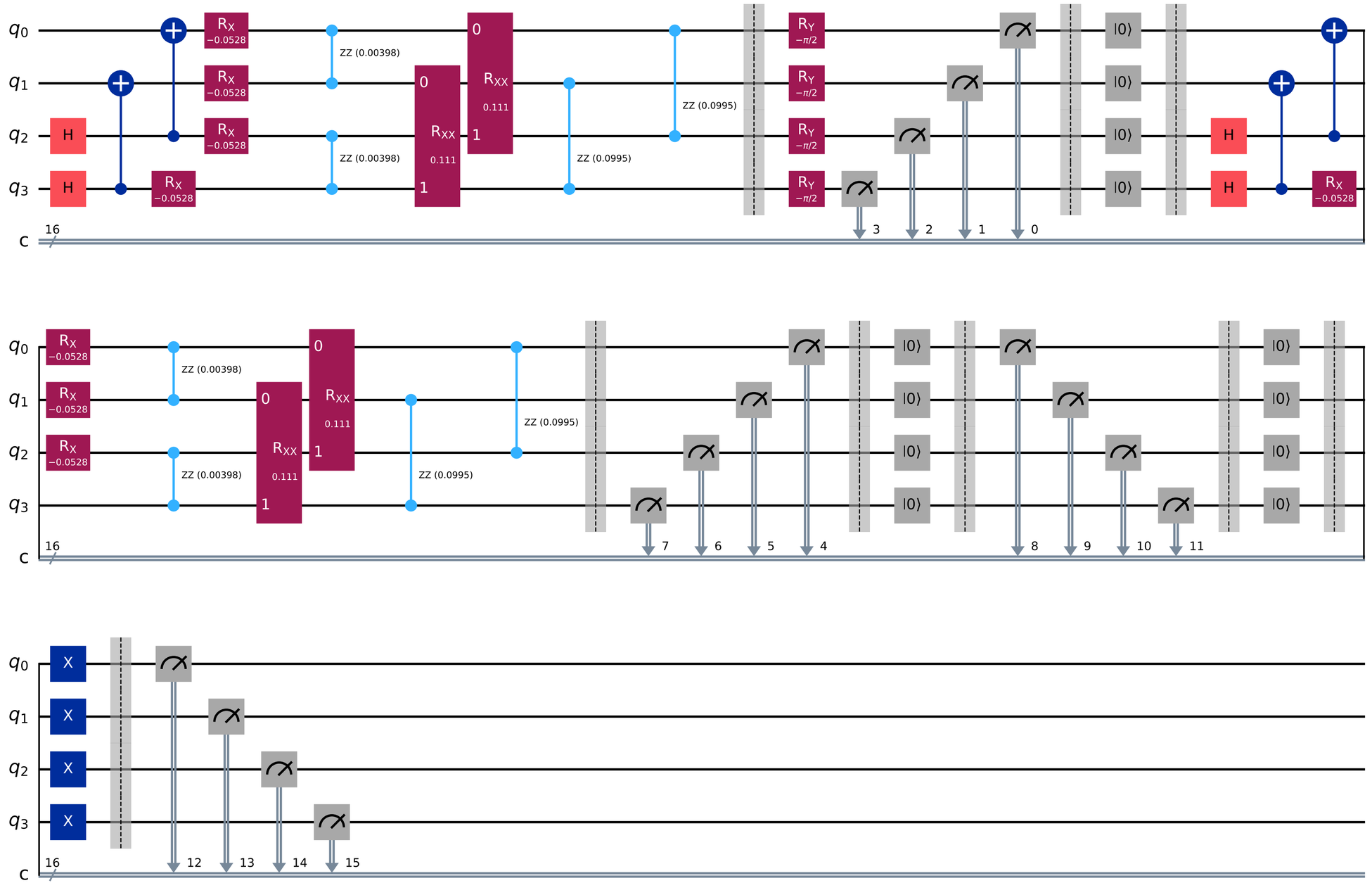

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.253
R_X −0.253
R_X −0.253
R_X −0.253
ZZ (−0.255)
ZZ (−0.255)
R_{XX} 0.0981
R_{XX} 0.0981
ZZ (0.0995)
ZZ (0.0995)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.253
R_X −0.253
R_X −0.253
R_X −0.253
3
2
1
0
ZZ (−0.255)
ZZ (−0.255)
R_{XX} 0.0981
R_{XX} 0.0981
ZZ (0.0995)
ZZ (0.0995)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15

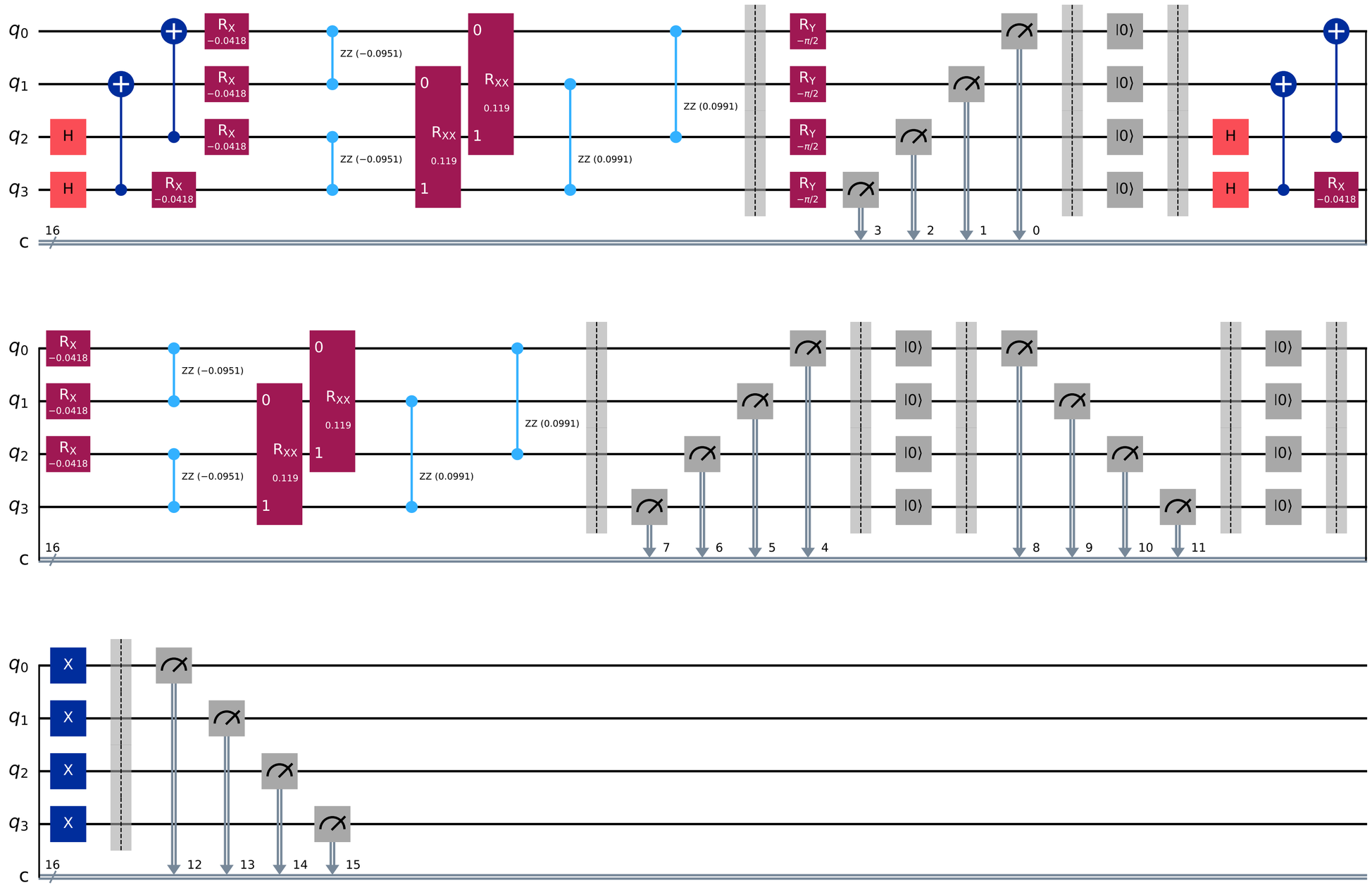

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0403
R_X −0.0403
R_X −0.0403
R_X −0.0403
ZZ (0.307)
ZZ (0.307)
R_{XX} 0.0992
ZZ (0.0998)
ZZ (0.0998)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0403
R_X −0.0403
R_X −0.0403
R_X −0.0403
ZZ (0.307)
ZZ (0.307)
R_{XX} 0.0992
ZZ (0.0998)
ZZ (0.0998)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

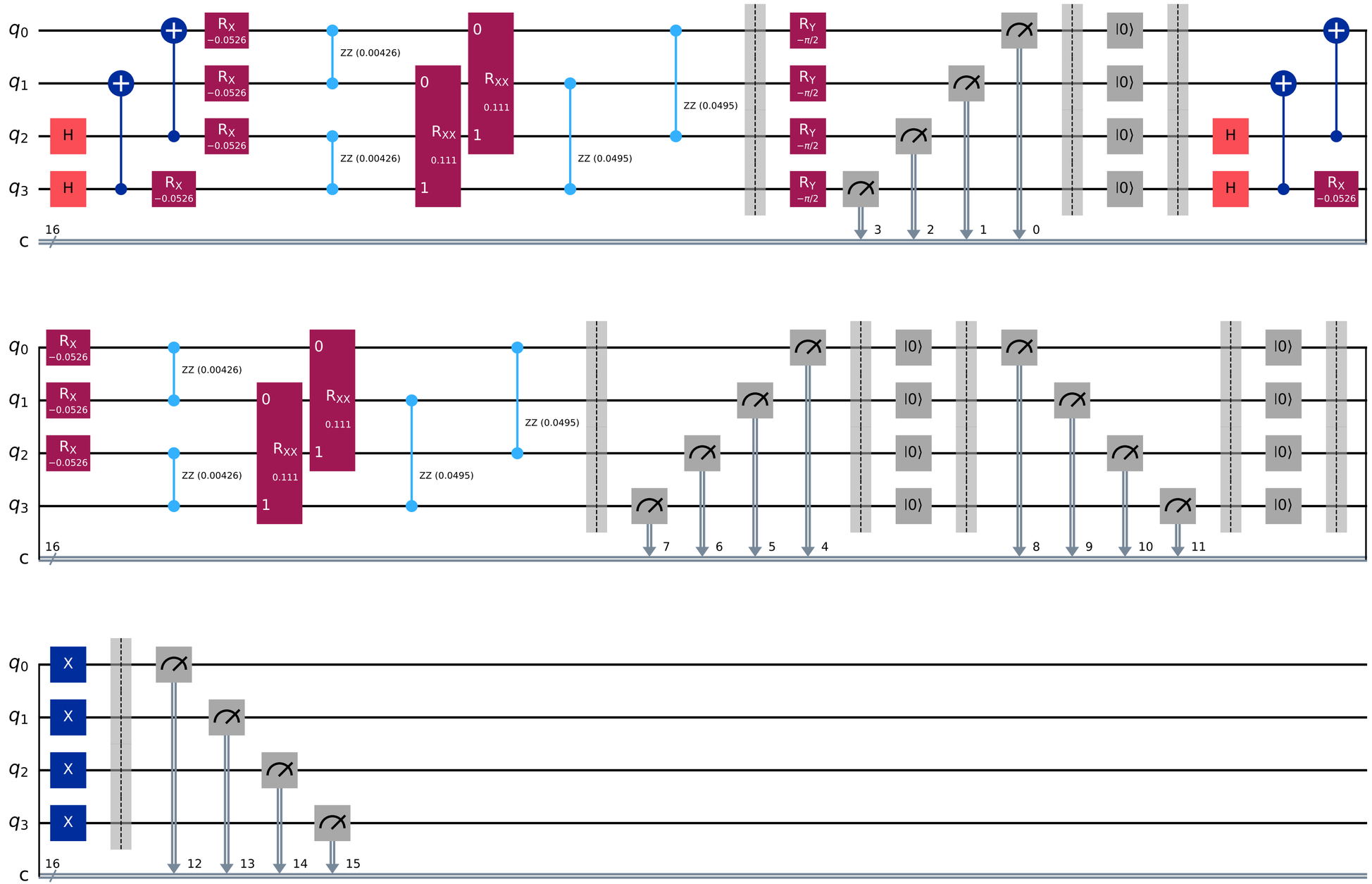

q0
q1
q2
q3
c
16
H
H
RX 0.197
RX 0.197
RX 0.197
RX 0.197
ZZ (0.0777)
ZZ (0.0777)
RXX 0.0992
ZZ (0.0998)
ZZ (0.0998)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.197
RX 0.197
RX 0.197
RX 0.197
ZZ (0.0777)
ZZ (0.0777)
RXX 0.0992
ZZ (0.0998)
ZZ (0.0998)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

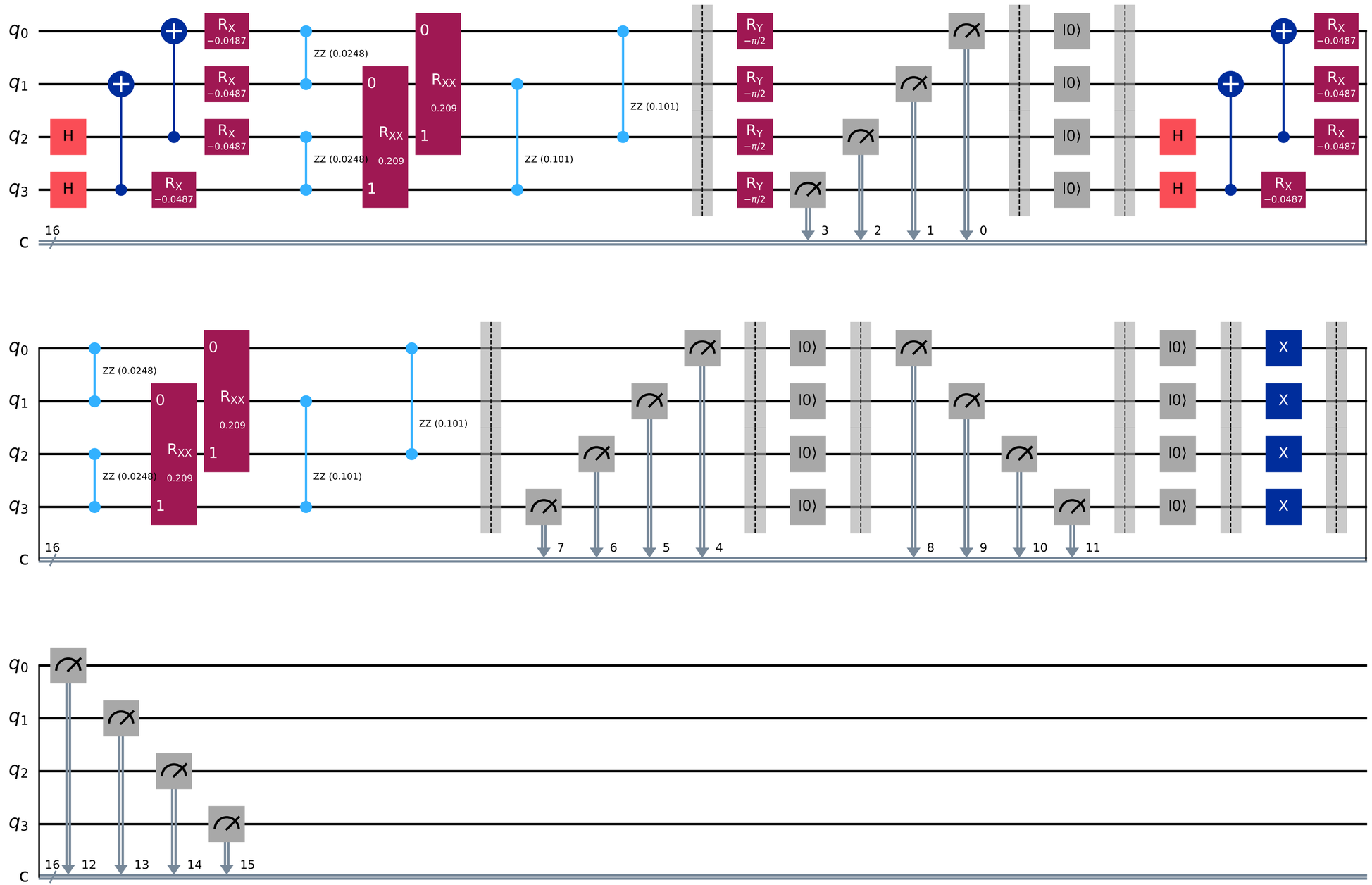

q0
q1
q2
q3
c
16
H
H
RX 0.0994
RX 0.0994
RX 0.0994
RX 0.0994
ZZ (0.0994)
ZZ (0.0994)
RXX 0.05
ZZ (0.1)
ZZ (0.1)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0994
RX 0.0994
RX 0.0994
RX 0.0994
ZZ (0.0994)
ZZ (0.0994)
RXX 0.05
ZZ (0.1)
ZZ (0.1)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

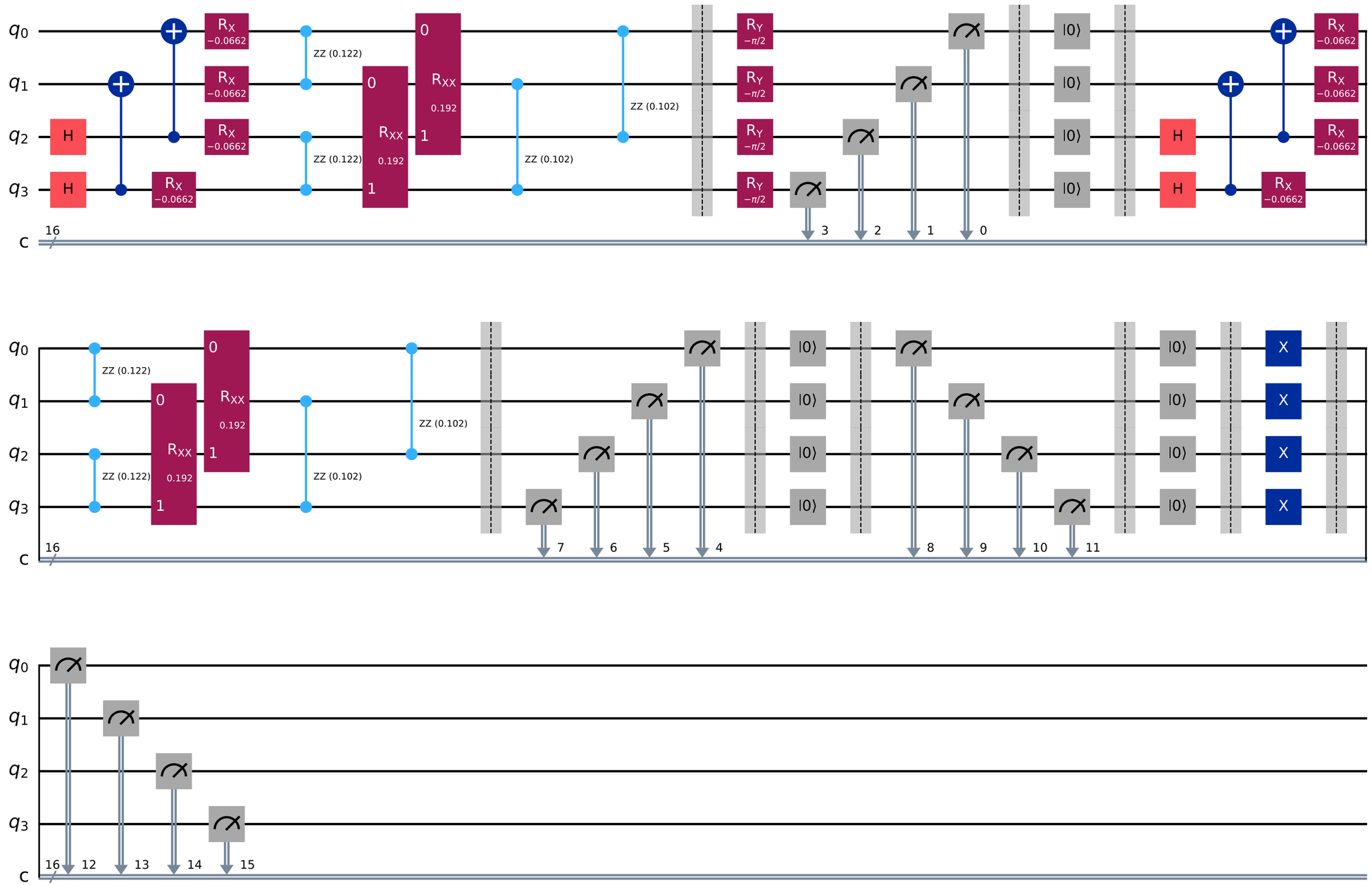

q0
q1
q2
q3
c
16
H
H
RX 0.0463
RX 0.0463
RX 0.0463
RX 0.0463
ZZ (0.0157)
ZZ (0.0157)
RXX 0.104
ZZ (0.0999)
ZZ (0.0999)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0463
RX 0.0463
RX 0.0463
RX 0.0463
ZZ (0.0157)
ZZ (0.0157)
RXX 0.104
RXX 0.104
ZZ (0.0999)
ZZ (0.0999)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

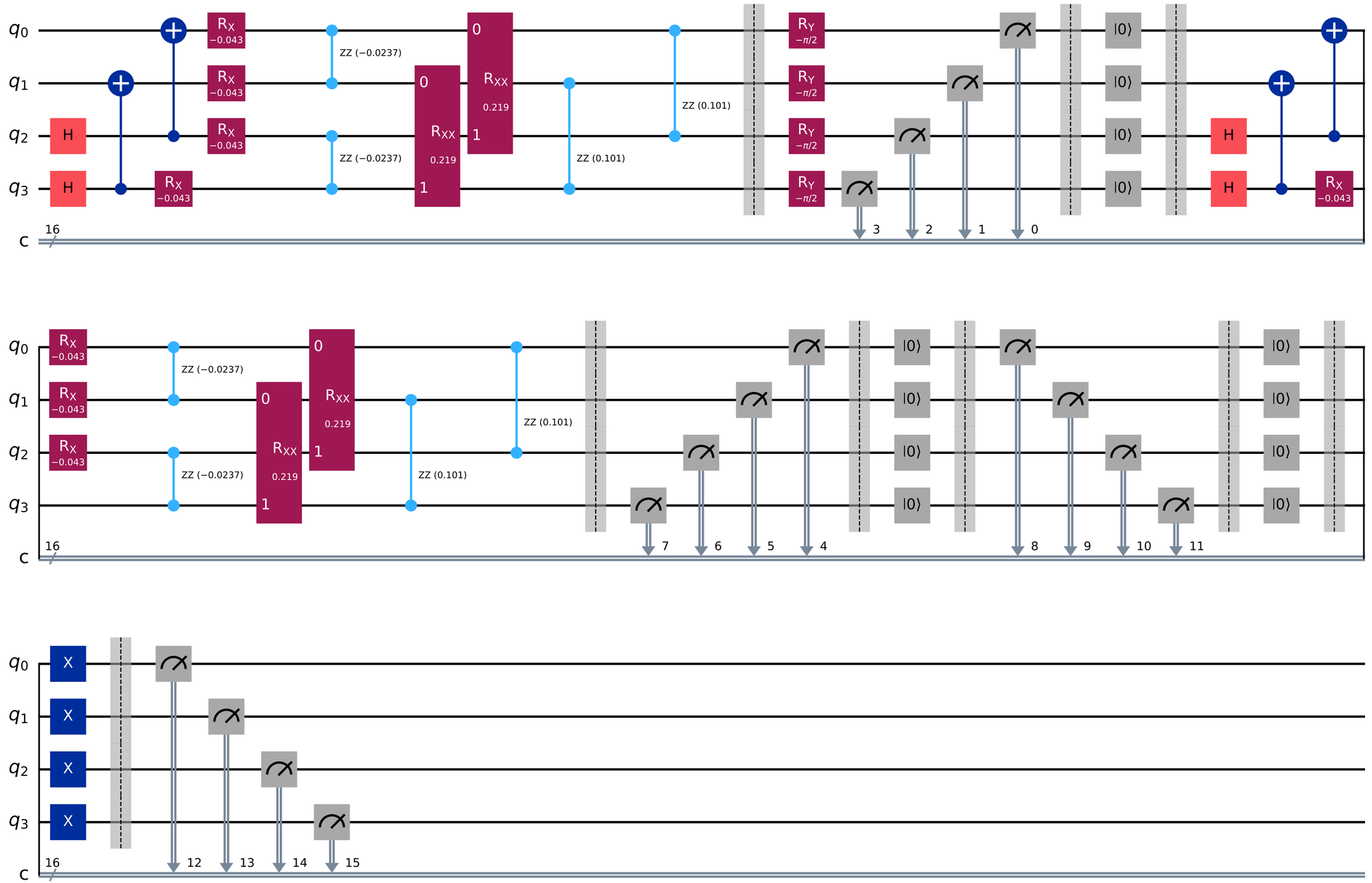

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0528
R_X −0.0528
R_X −0.0528
R_X −0.0528
ZZ (0.00398)
ZZ (0.00398)
0
R_{XX}
0.111
1
0
R_{XX}
0.111
1
ZZ (0.0995)
ZZ (0.0995)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0528
3
2
1
0
R_X −0.0528
R_X −0.0528
R_X −0.0528
ZZ (0.00398)
ZZ (0.00398)
0
R_{XX}
0.111
1
ZZ (0.0995)
ZZ (0.0995)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

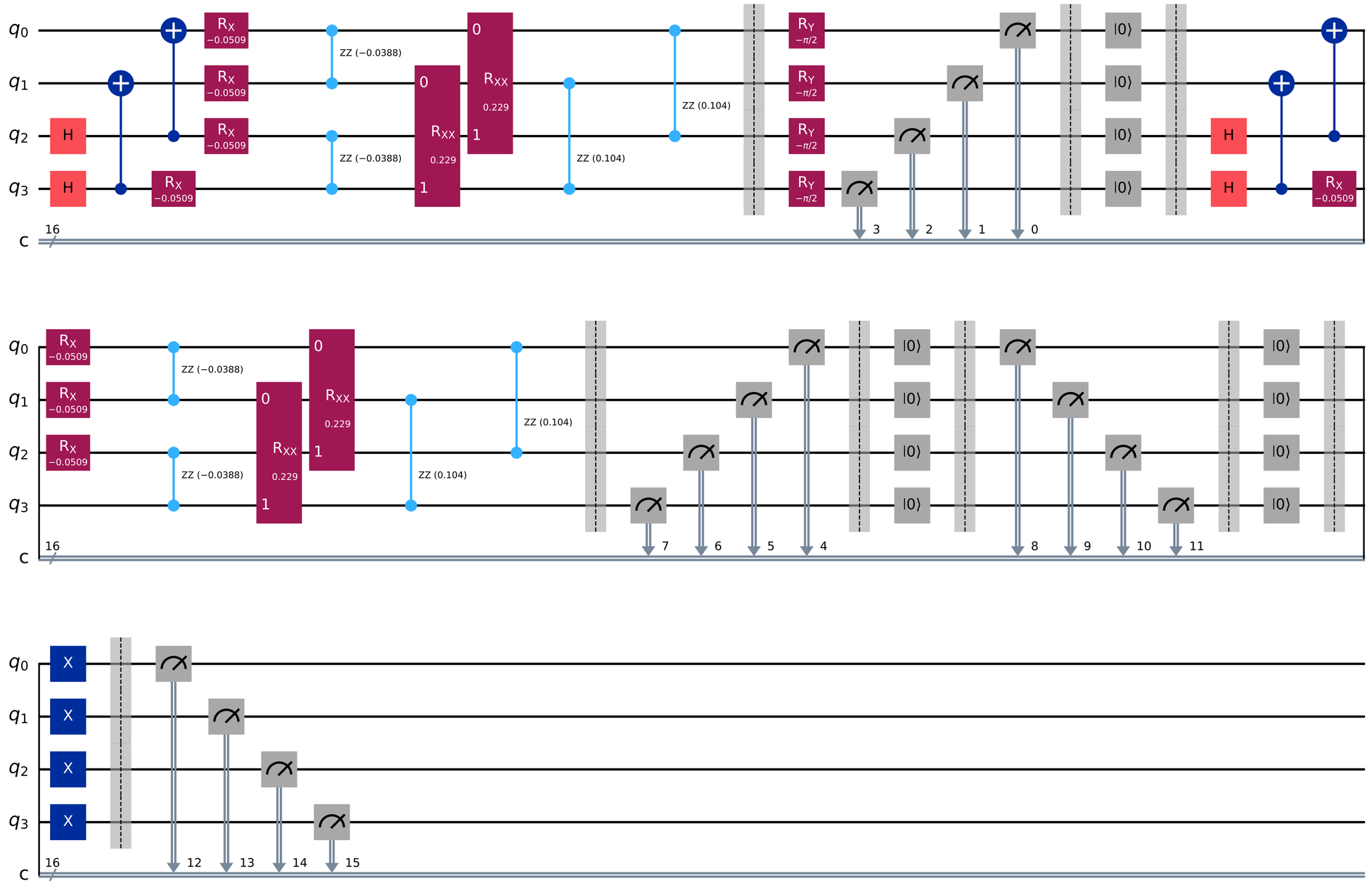

q0
q1
q2
q3
c
16
H
H
RX −0.0418
RX −0.0418
RX −0.0418
RX −0.0418
ZZ (−0.0951)
ZZ (−0.0951)
0
RXX
0.119
1
0
RXX
0.119
1
ZZ (0.0991)
ZZ (0.0991)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0418
3
2
1
0
RX −0.0418
RX −0.0418
RX −0.0418
ZZ (−0.0951)
ZZ (−0.0951)
0
RXX
0.119
1
ZZ (0.0991)
ZZ (0.0991)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

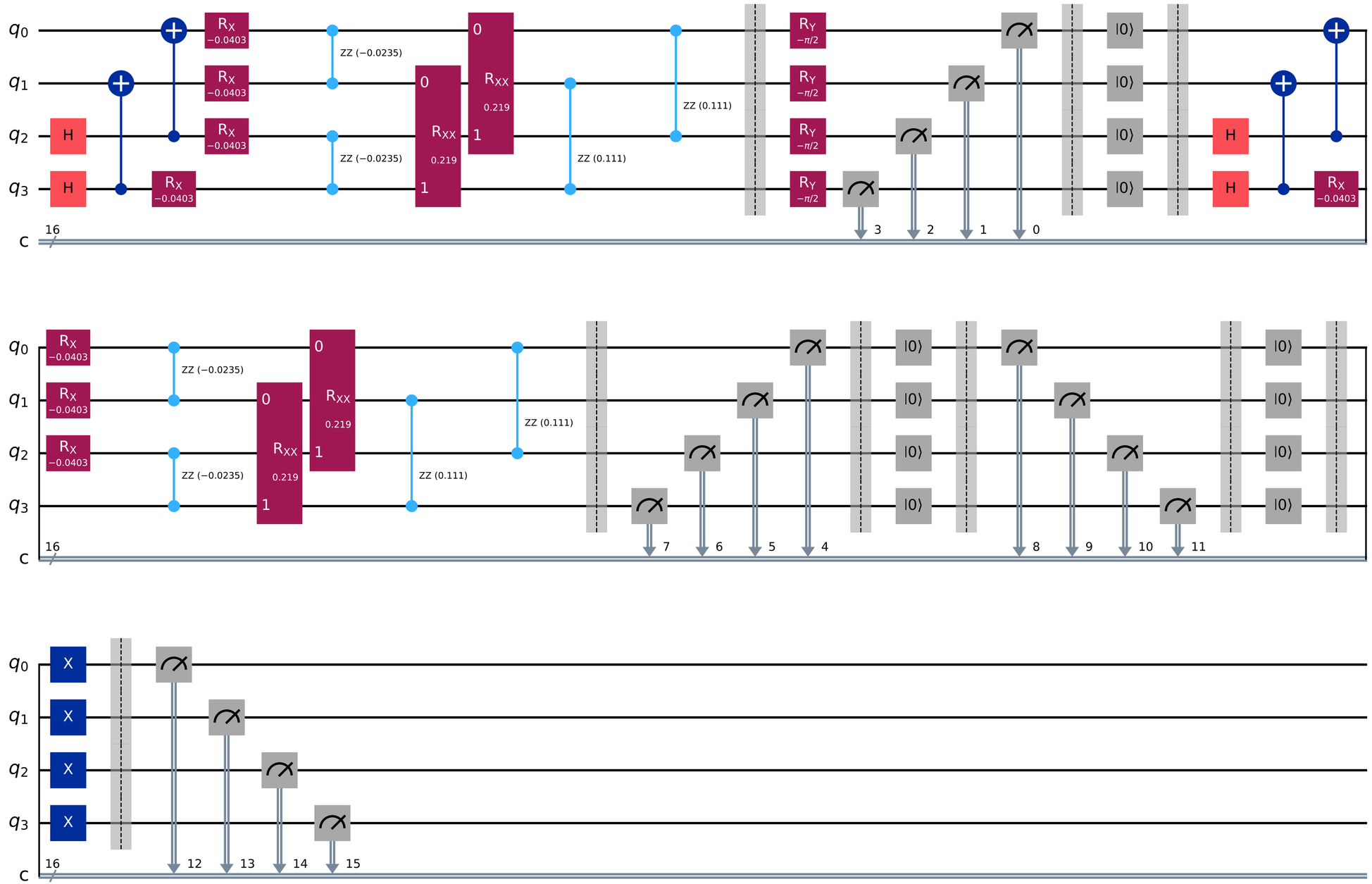

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0526
R_Y −π/2
ZZ (0.00426)
ZZ (0.0495)
R_{XX} 0.111
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0487
R_X −0.0487
R_X −0.0487
R_X −0.0487
ZZ (0.0248)
ZZ (0.0248)
0
R_{XX}
0.209
1
0
R_{XX}
0.209
1
ZZ (0.101)
ZZ (0.101)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0487
R_X −0.0487
R_X −0.0487
R_X −0.0487
ZZ (0.0248)
ZZ (0.0248)
0
R_{XX}
0.209
1
ZZ (0.101)
ZZ (0.101)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
q_0
q_1
q_2
q_3
c
16
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.0662
RX −0.0662
RX −0.0662
RX −0.0662
ZZ (0.122)
ZZ (0.122)
0
RXX
0.192
1
ZZ (0.102)
ZZ (0.102)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0662
RX −0.0662
RX −0.0662
RX −0.0662
ZZ (0.122)
ZZ (0.122)
ZZ (0.102)
ZZ (0.102)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

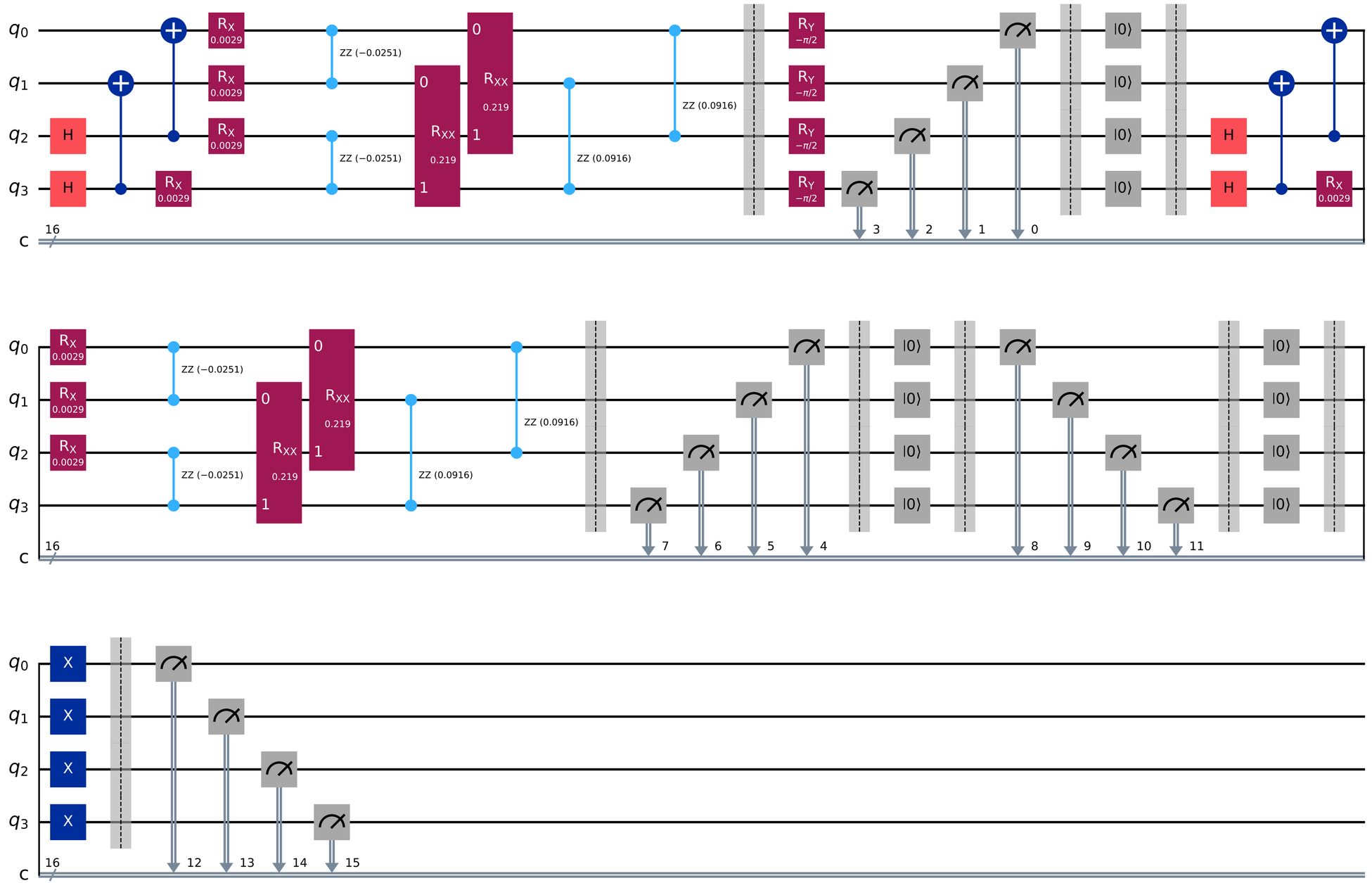

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.043
ZZ (−0.0237)
ZZ (0.101)
R_{XX} 0.219
R_Y −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

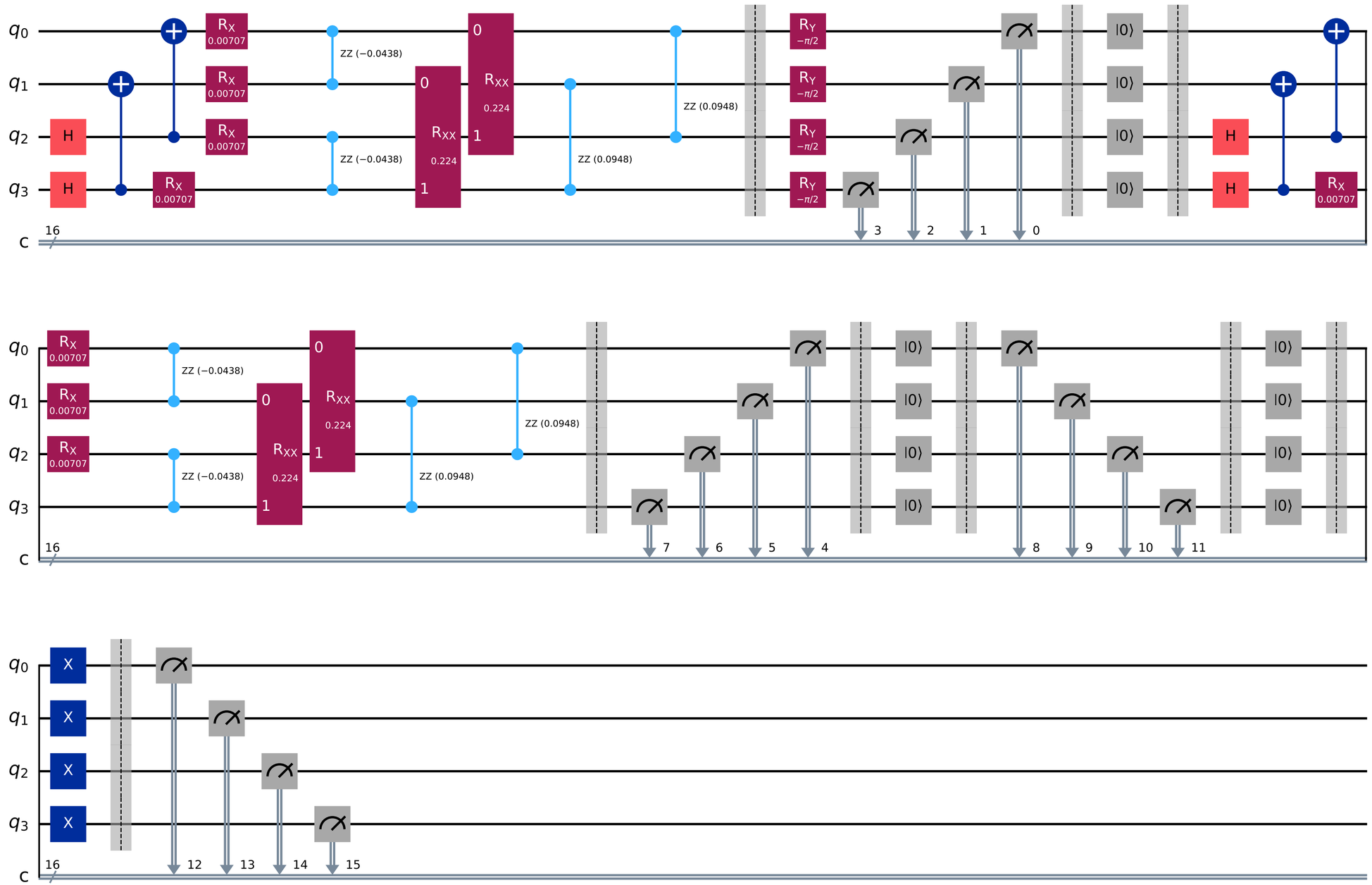

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0509
ZZ (−0.0388)
ZZ (−0.0388)
0
R_{XX}
0.229
1
ZZ (0.104)
ZZ (0.104)
R_Y −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

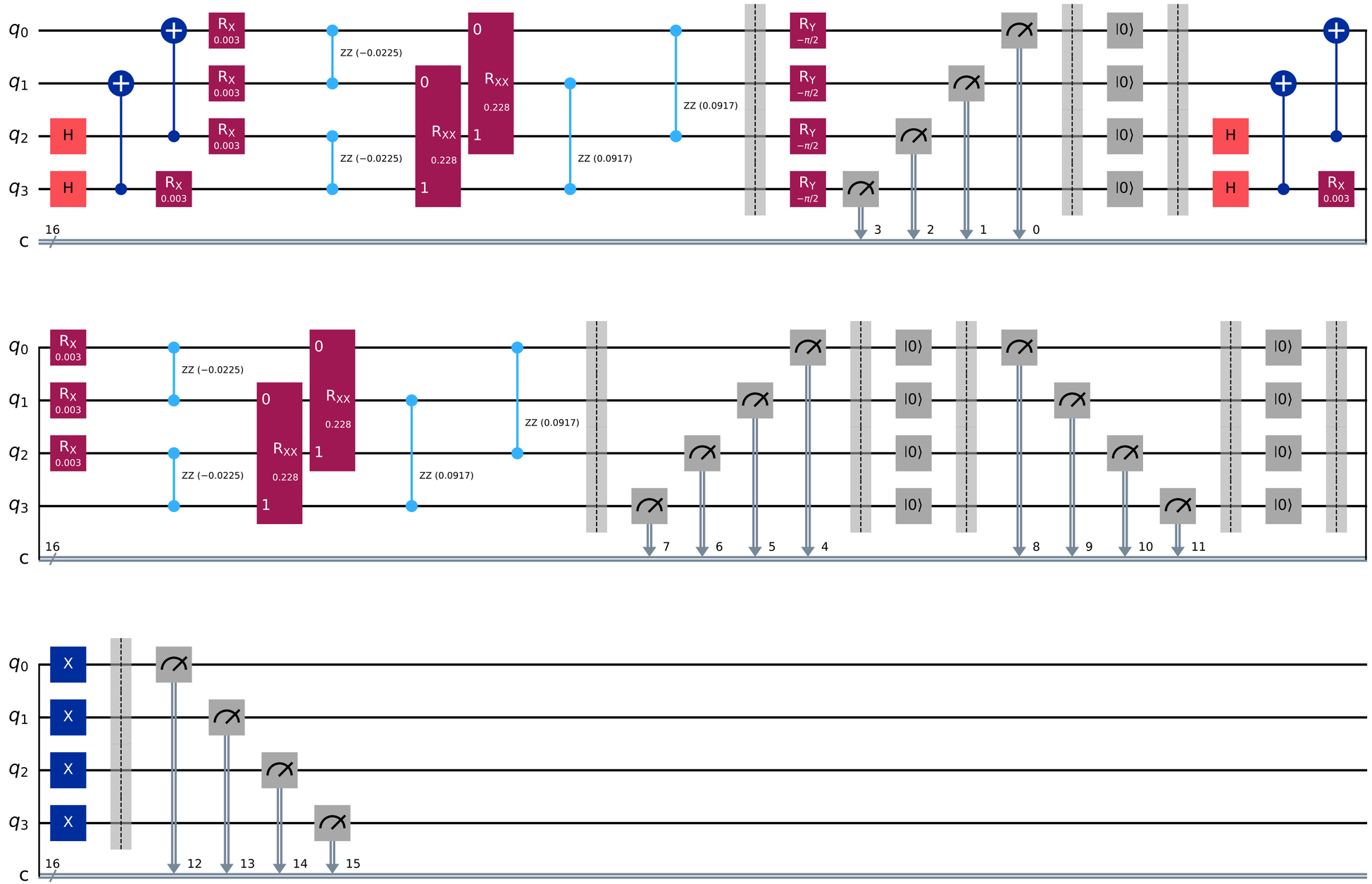

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0403
R_X −0.0403
R_X −0.0403
R_X −0.0403
ZZ (−0.0235)
ZZ (−0.0235)
0
R_{XX}
0.219
1
0
R_{XX}
0.219
1
ZZ (0.111)
ZZ (0.111)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0403
3
2
1
0
R_X −0.0403
R_X −0.0403
R_X −0.0403
ZZ (−0.0235)
ZZ (−0.0235)
0
R_{XX}
0.219
1
ZZ (0.111)
ZZ (0.111)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

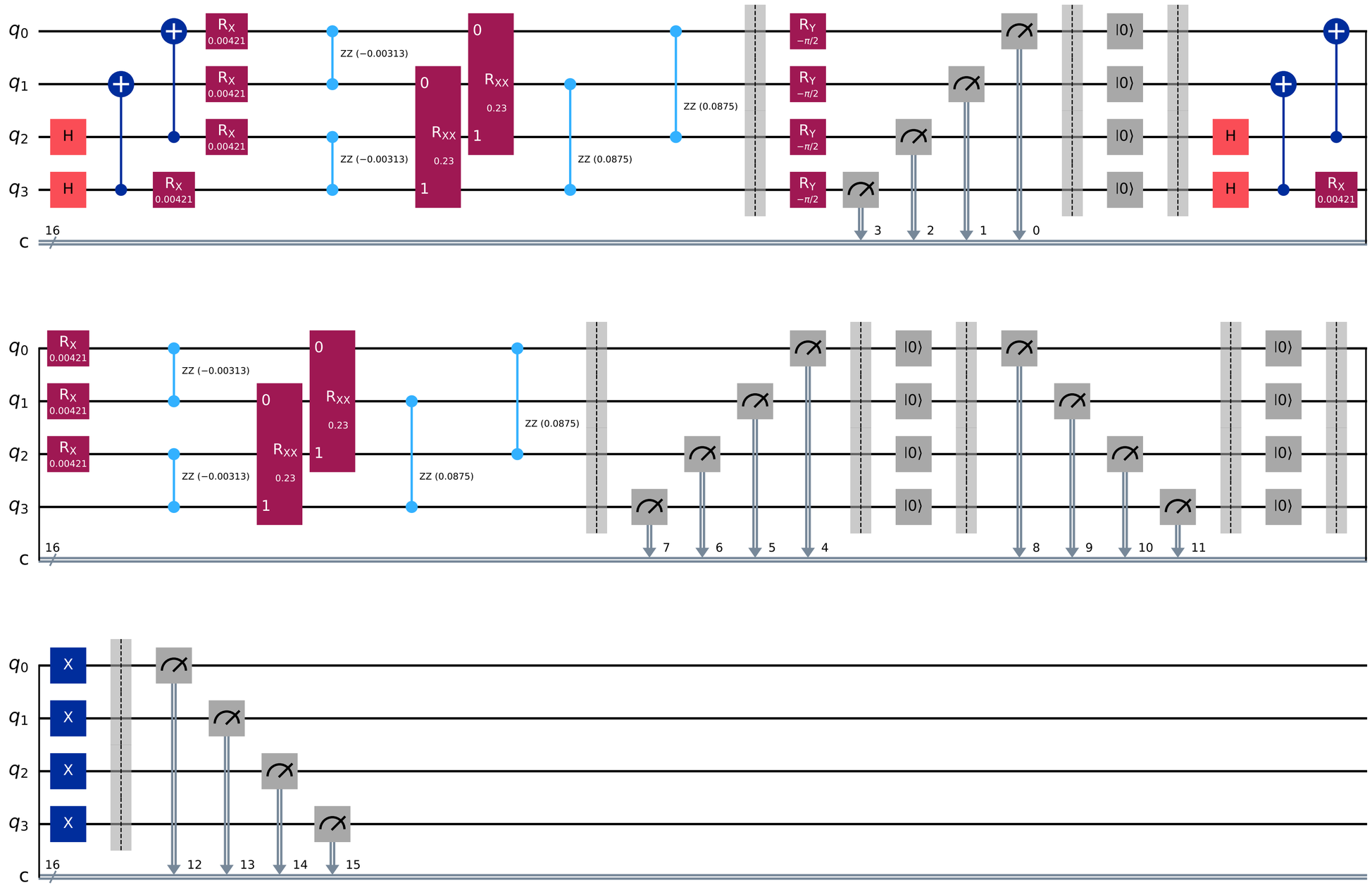
q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0205
ZZ (−0.0219)
ZZ (−0.0219)
0
R_{XX}
0.218
1
ZZ (0.108)
ZZ (0.108)
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

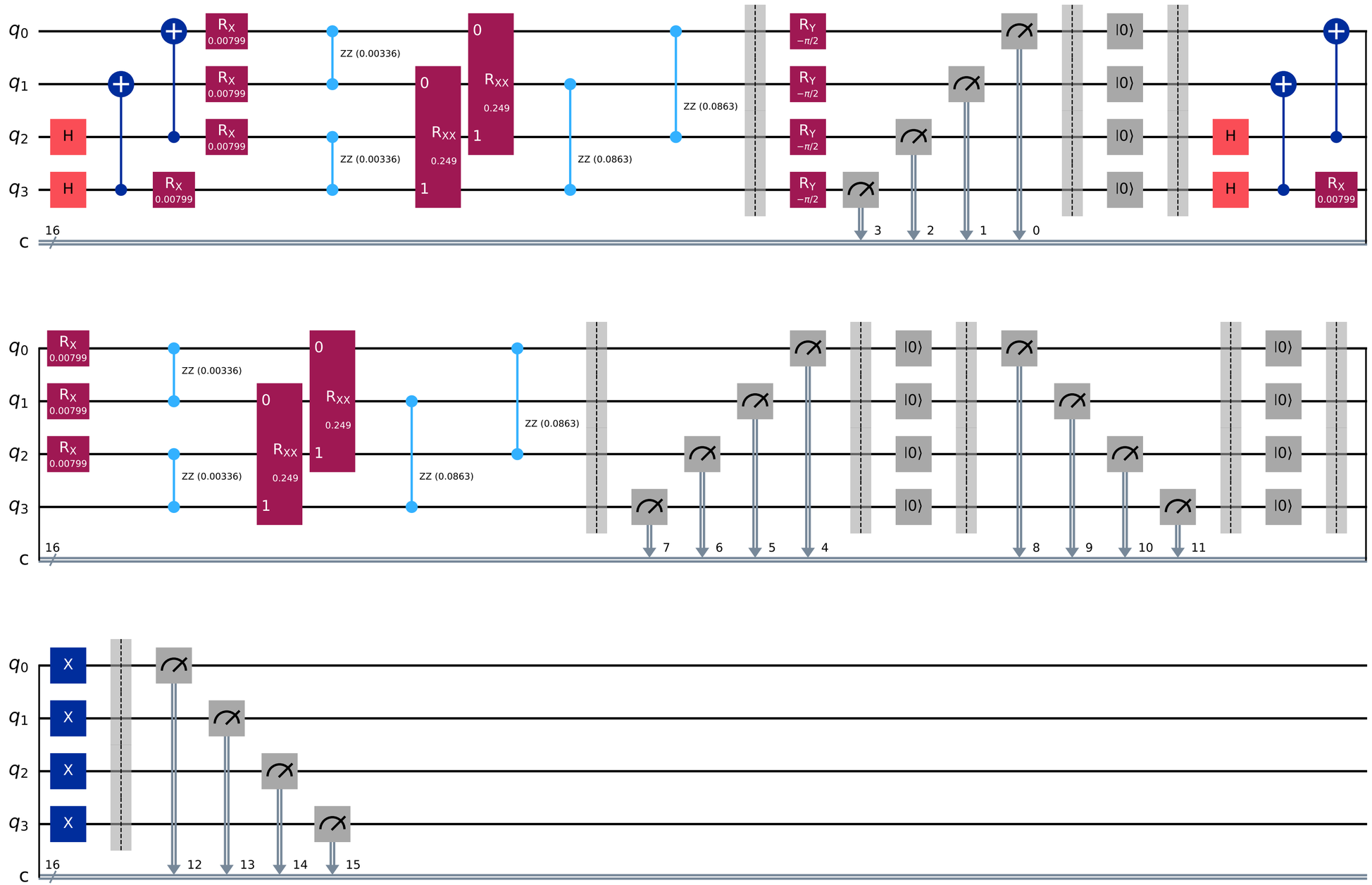
q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.000759
R_X −0.000759
R_X −0.000759
R_X −0.000759
ZZ (−0.0212)
ZZ (−0.0212)
0
R_{XX}
0.218
1
0
R_{XX}
0.218
1
ZZ (0.111)
ZZ (0.111)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0029
R_X 0.0029
R_X 0.0029
R_X 0.0029
ZZ (−0.0251)
ZZ (−0.0251)
R_{XX} 0.219
R_{XX} 0.219
ZZ (0.0916)
ZZ (0.0916)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.0029
3
2
1
0
R_X 0.0029
R_X 0.0029
R_X 0.0029
ZZ (−0.0251)
ZZ (−0.0251)
R_{XX} 0.219
R_{XX} 0.219
ZZ (0.0916)
ZZ (0.0916)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00707
ZZ (−0.0438)
ZZ (−0.0438)
R_{XX} 0.224
ZZ (0.0948)
ZZ (0.0948)
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.003
R_X 0.003
R_X 0.003
R_X 0.003
ZZ (−0.0225)
ZZ (−0.0225)
0
R_{XX} 0.228
1
0
R_{XX} 0.228
1
ZZ (0.0917)
ZZ (0.0917)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.003
3
2
1
0
R_X 0.003
R_X 0.003
R_X 0.003
ZZ (−0.0225)
ZZ (−0.0225)
0
R_{XX} 0.228
1
0
R_{XX} 0.228
1
ZZ (0.0917)
ZZ (0.0917)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00421
ZZ (−0.00313)
ZZ (−0.00313)
0
R_{XX} 0.23
1
ZZ (0.0875)
ZZ (0.0875)
R_Y $-\pi/2$
|0⟩
3
2
1
0
X
7
6
5
4
8
9
10
11
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX 0.00799
ZZ (0.00336)
ZZ (0.00336)
RXX 0.249
ZZ (0.0863)
ZZ (0.0863)
RY −π/2
3
2
1
0
|0⟩
H
H
RX 0.00799
RX 0.00799
ZZ (0.00336)
ZZ (0.00336)
RXX 0.249
ZZ (0.0863)
ZZ (0.0863)
7
6
5
4
8
9
10
11
|0⟩
X
X
X
X
12
13
14
15